CellRanger cell calling

陈明

两种cell calling的算法

It uses a cutoff based on total UMI counts for each barcode to identify cells. This step identifies the initial number of high RNA content cells (initial cells).

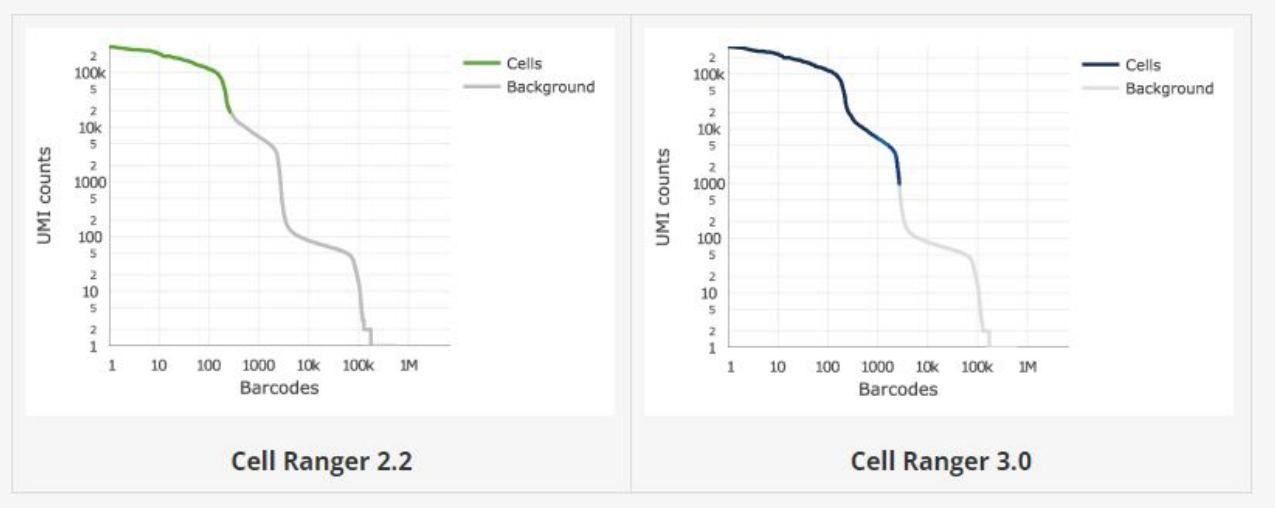
1. 计算 expected cell(N)
2. 计算Top N的99分位数 m
3. m / 10作为是否是细胞的阈值

两种cell calling的算法

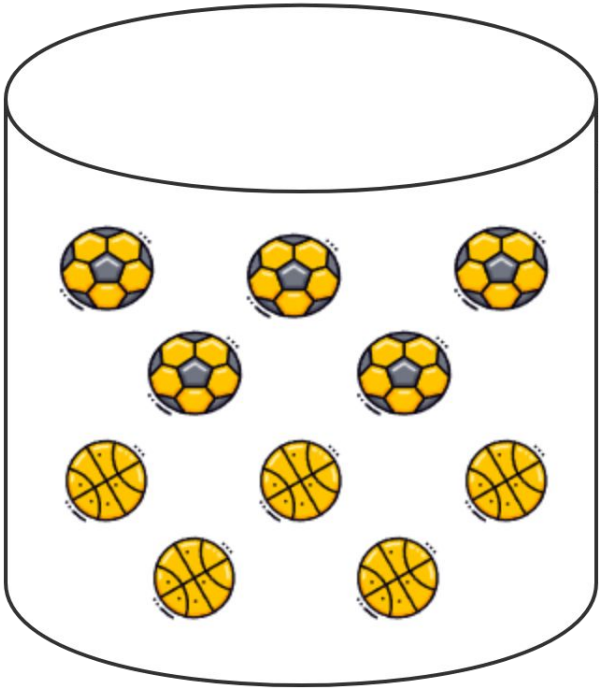
Then the algorithm uses the RNA profile of each remaining barcode to determine if it is an "empty" or a cell-containing partition. This second step captures low RNA content cells whose total UMI counts may be similar to empty GEMs.

1. 获取一些背景的Barcodes并构建模型
2. 将模型与剩余未鉴定未细胞的Barcodes进行比较，将与模型表达谱差异大的细胞作为细胞call出

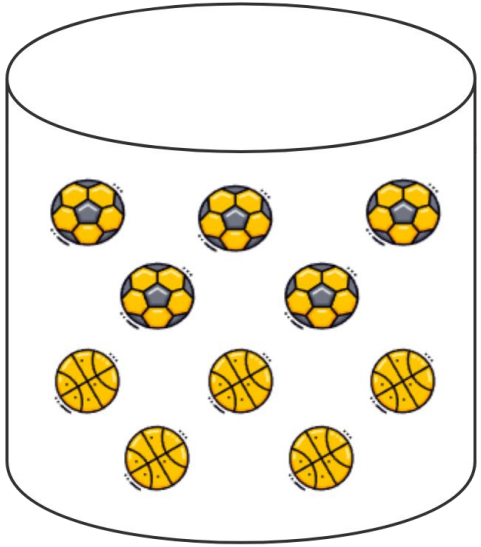
两种cell calling的算法

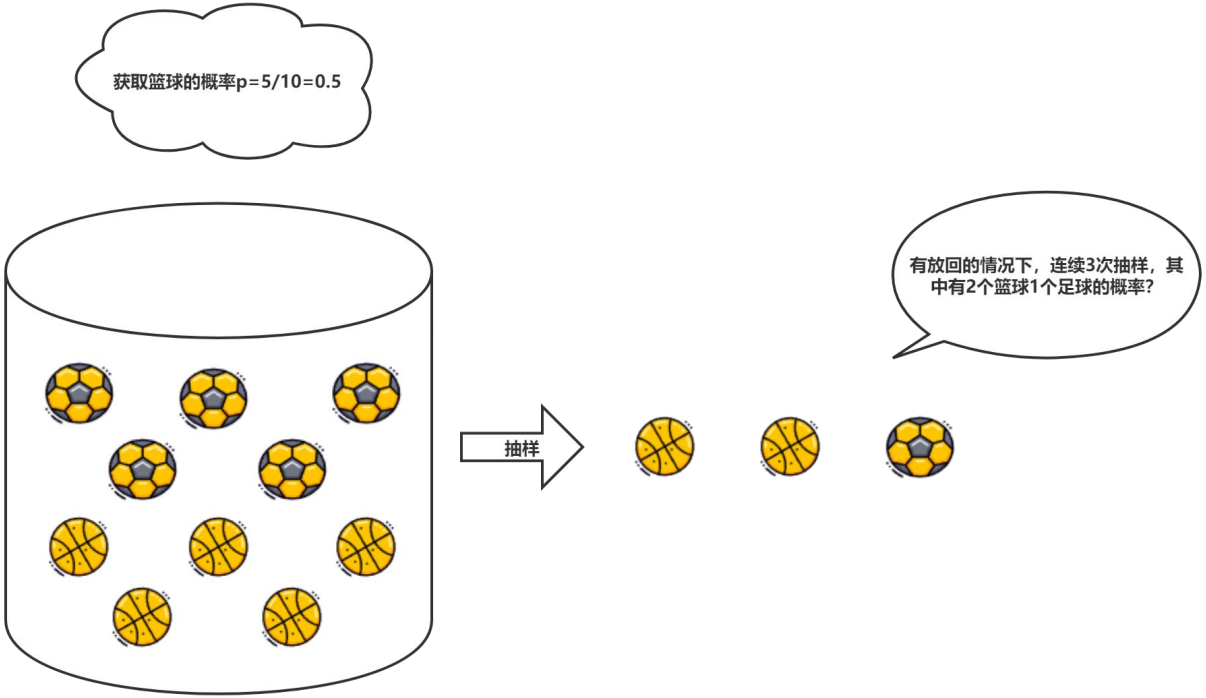


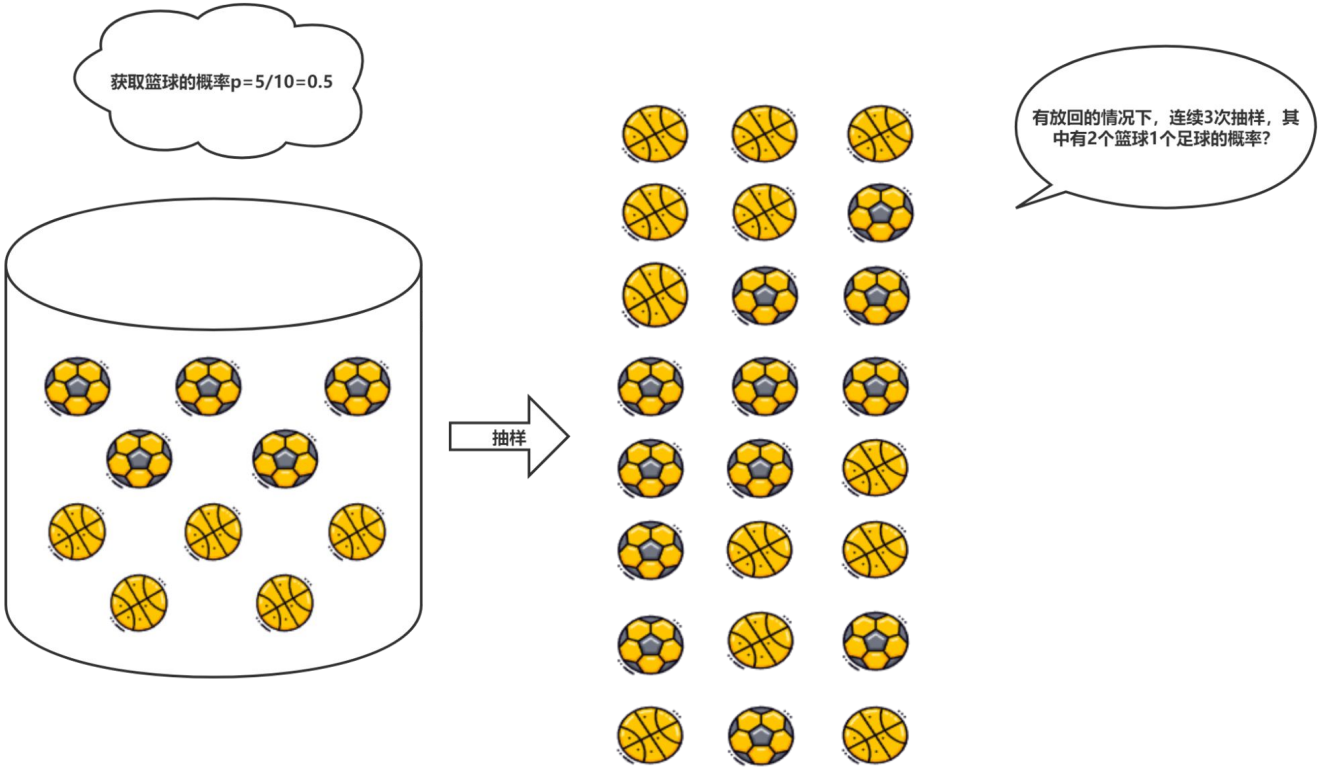
第二种cell calling算法是如何实现的？



获取篮球的概率p=5/10=0.5




获取篮球的概率p=5/10=0.5
抽样
有放回的情况下，连续3次抽样，其中有2个篮球1个足球的概率?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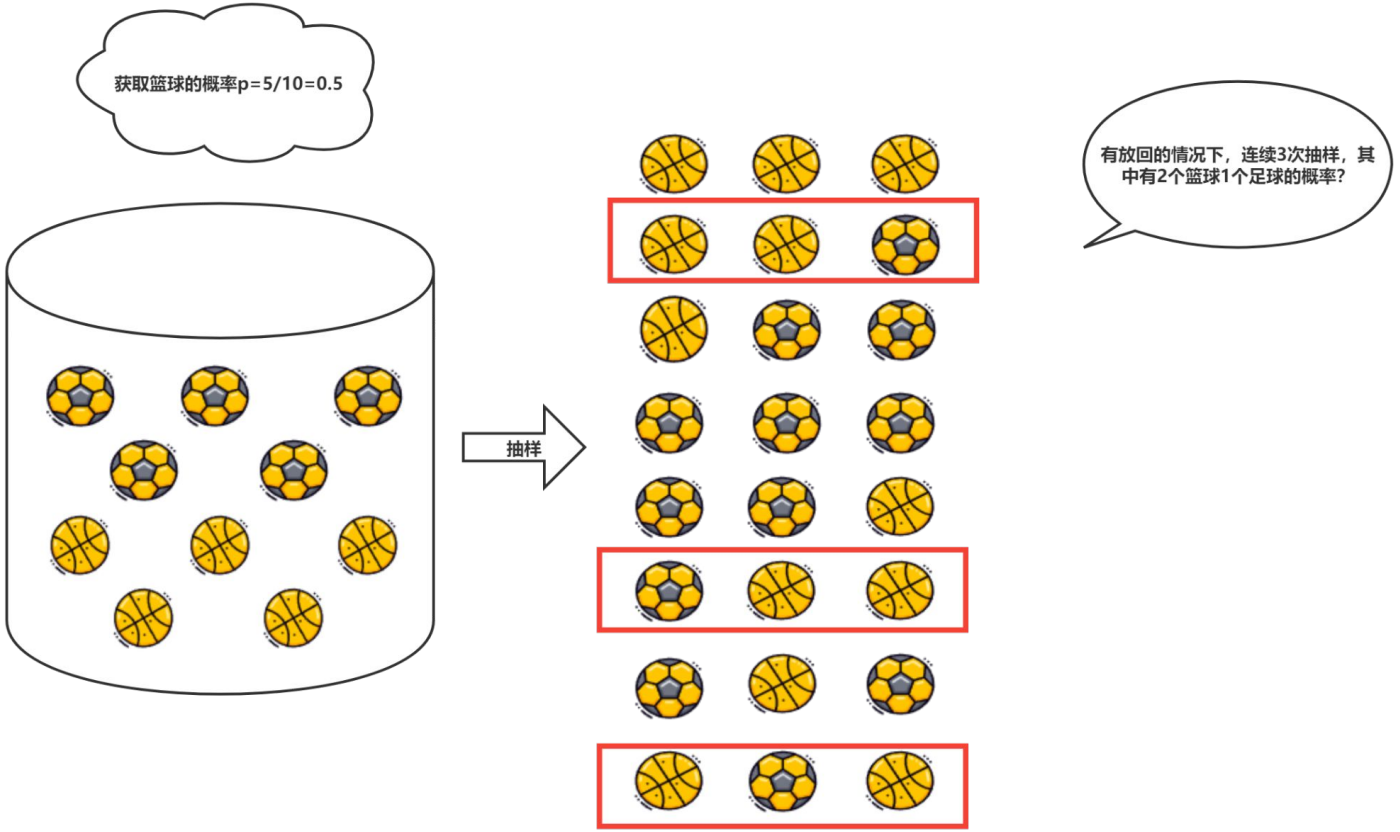
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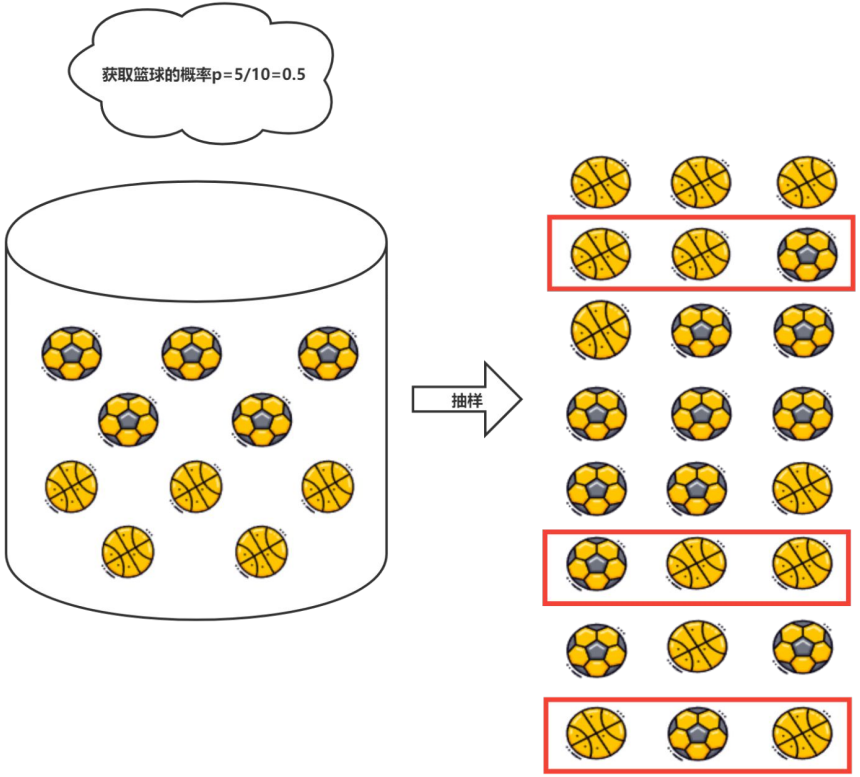
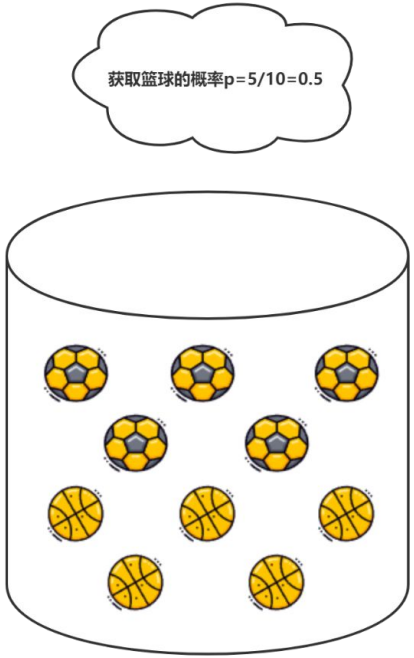


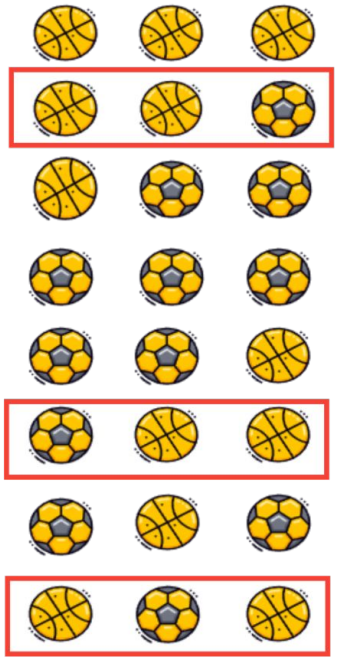
有放回的情况下，连续3次抽样，其中有2个篮球1个足球的概率？

$$p(2) = \frac{3}{8} = 0.375$$

二项分布 $f(k,n,p) = \binom{n}{k} p^k (1-p)^{n-k}$



抽样



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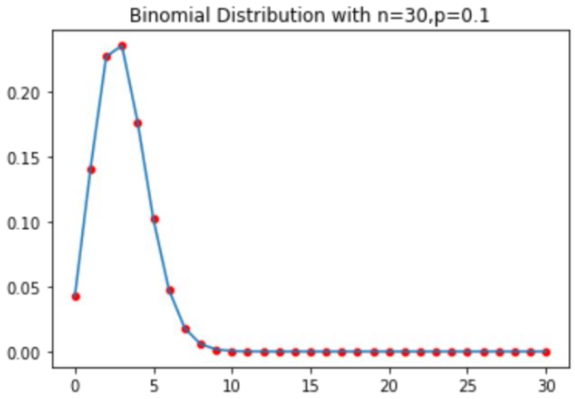
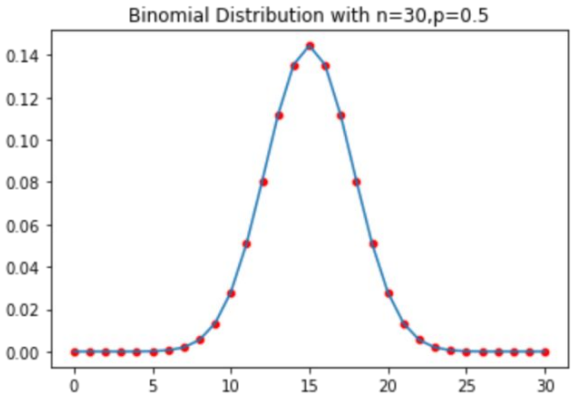
$$p(2) = \frac{3}{8} = 0.375$$

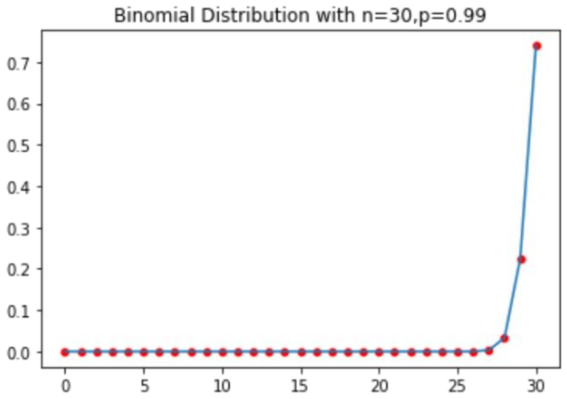
二项分布PMF(Probability mass function)

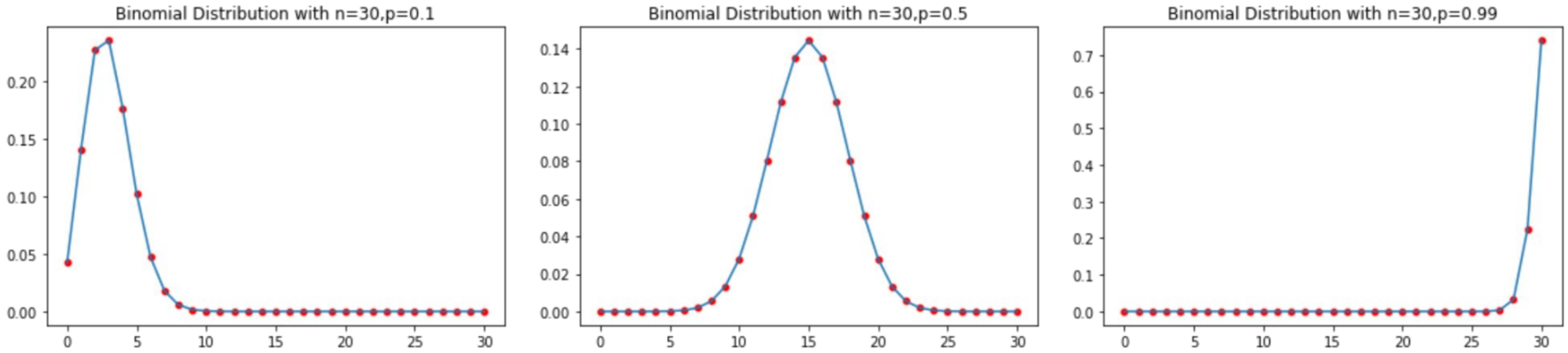
$$f(k, n, p) = \binom{n}{k} p^k (1-p)^{n-k}$$

$$p(2) = \binom{3}{2} 0.5^2 (1-0.5)^{3-2} = 0.375$$


Binomial Distribution with n=30,p=0.1
0.20
0.15
0.10
0.05
0.00
0
5
10
15
20
25
30


Binomial Distribution with n=30,p=0.5
0.14
0.12
0.10
0.08
0.06
0.04
0.02
0.00
0
5
10
15
20
25
30

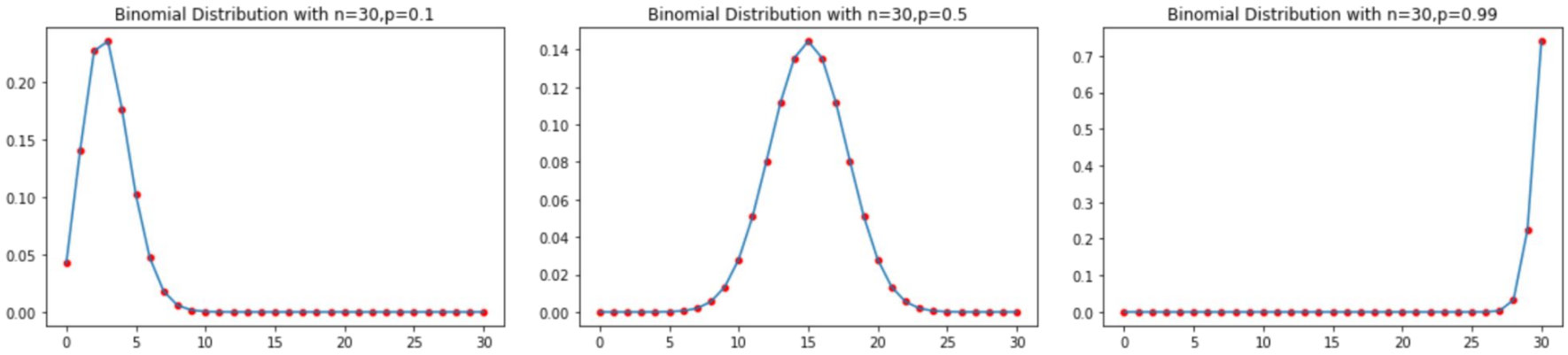

Binomial Distribution with n=30,p=0.99
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
5
10
15
20
25
30



如果某个事件(事件A)发生的概率是0.001，那么该事件发生时我们应该感到意外吗?

全集 = sum([0.001,0.001,....0.001]) = 1

全集 = sum([0.001,0.099,0.1,....0.1]) = 1



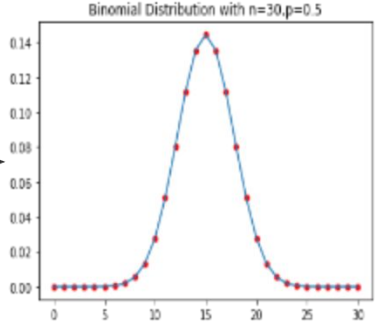
如果某个事件(事件A)发生的概率是0.001，那么该事件在所有可能的事件中是小概率事件吗？(显著性水平假定为0.01)

使用Monta Carlo simulation找到答案

通过大量随机抽样找到接近正确的答案

Monte Carlo Simulation分析思路

生成随机离散变量X(随机事件)



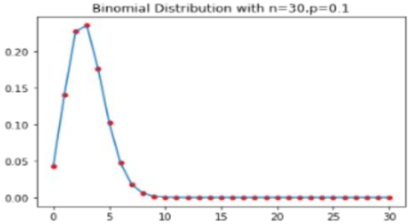
大量重复抽样(例如1,000,000次)

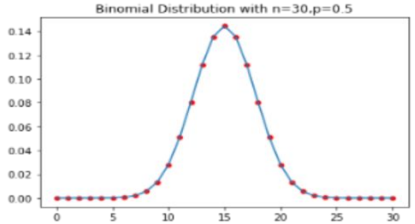
判断在所有的抽样事件中，该特定事件是否为小概率事件

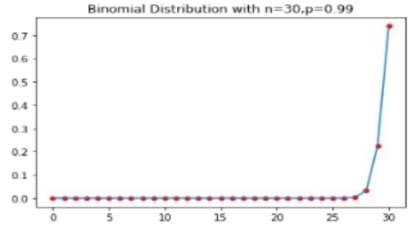
例1: 1,000,000次抽样中，99%的抽样结果的概率大于0.001,那么我们可以认为事件A在所有事件中是小概率事件

例2: 购买彩票1,000,000次，只中过1块钱,那么我们可以认为中奖是小概率事件

如果某个事件(事件A)发生的概率是0.001，那么该事件在所有可能的事件中是小概率事件吗？(显著性水平假定为0.01)







随机抽样1,000,000次并计算所有事件中比0.001概率更低的事件的比例

0.978628

0.001403

0.999997

是否小于0.01

否

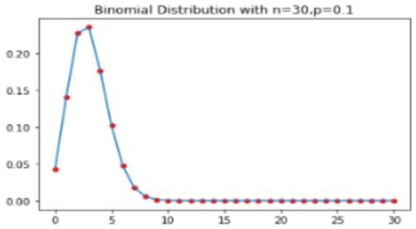
事件A是非小概率事件

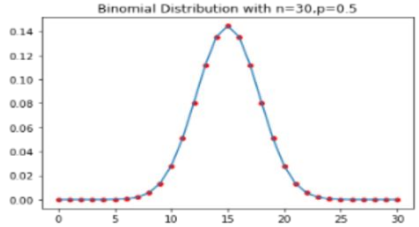
是

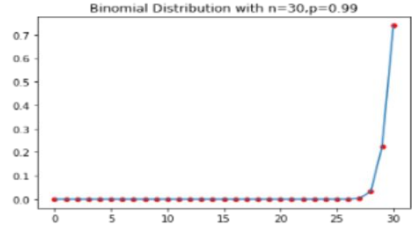
事件A是小概率事件

如果某个事件(事件A)发生的概率是0.001，那么该事件在所有可能的事件中是小概率事件吗？(显著性水平假定为0.01)

Monte Carlo Simulation让我们不用担心不同分布下的情况，而是借助计算机模拟快速找到一个接近正确的答案







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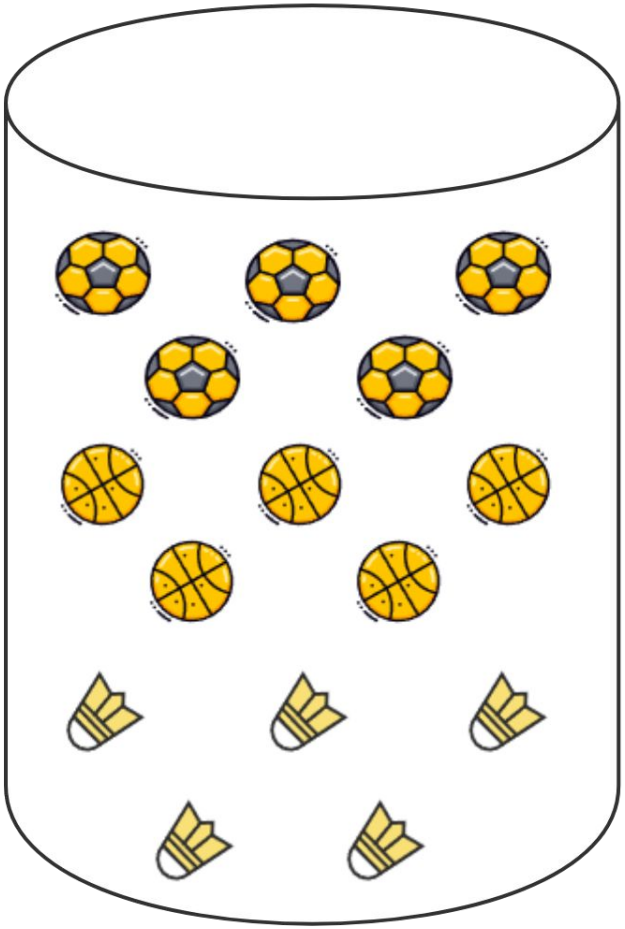
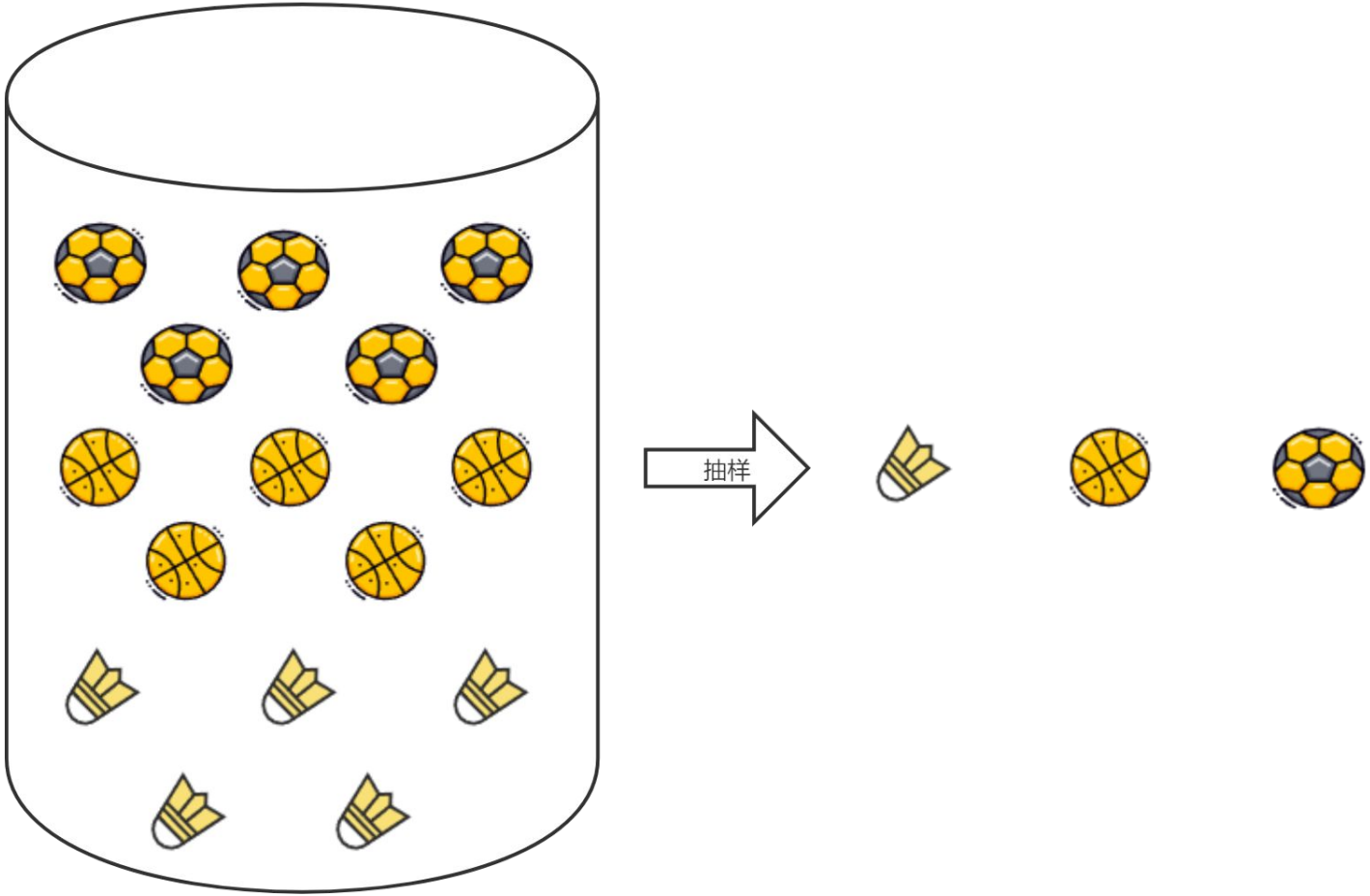
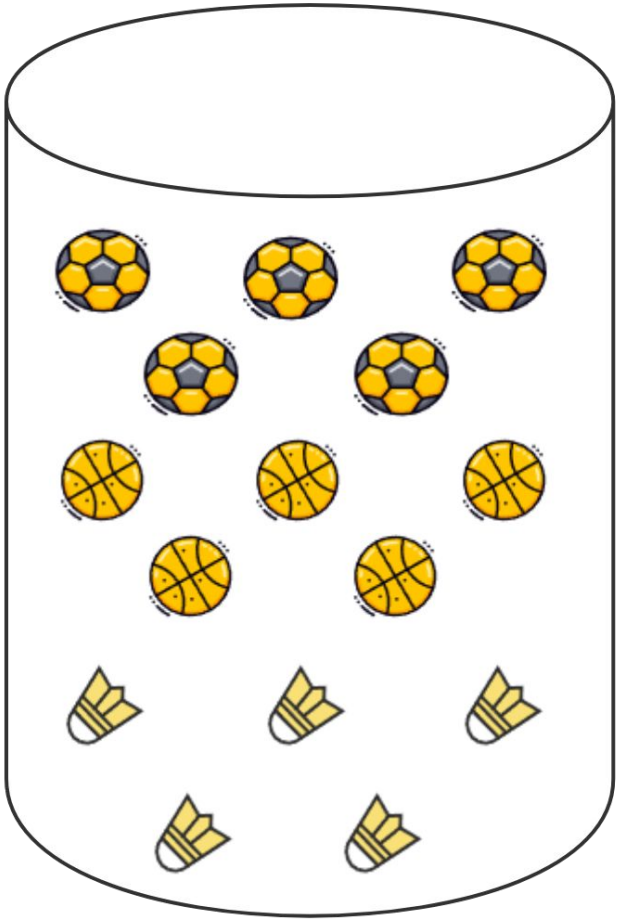


情况变得更为复杂，桶中增加了羽毛球


抽样

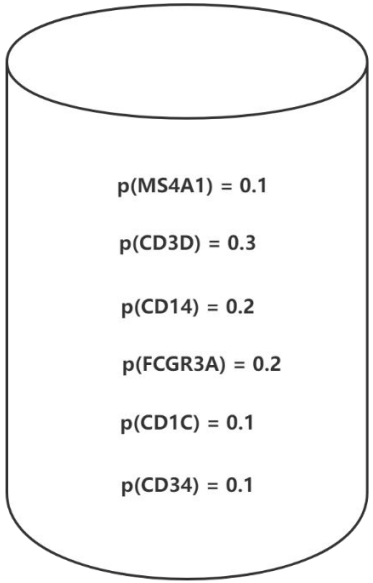


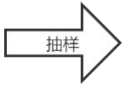
抽样

多项式PMF

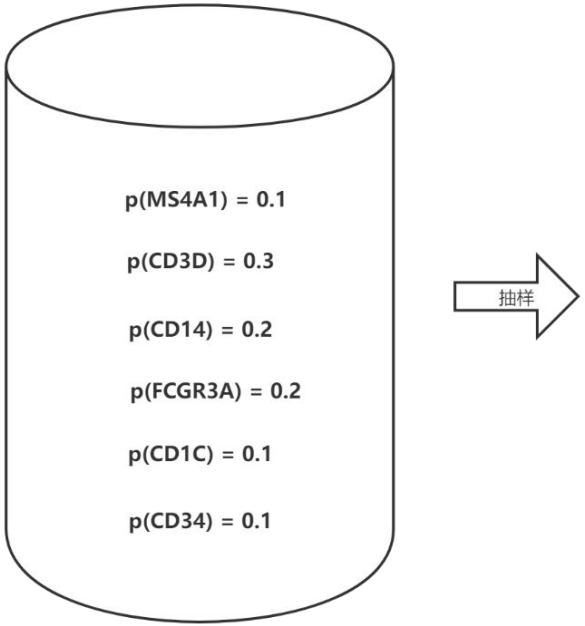
$$f(x) = \frac{n!}{x_1!\ldots x_k!} p_1^{x_1} \ldots p_k^{x_k}$$

$$p = \frac{3!}{1!1!1!} 1/3 * 1/3 * 1/3 = 0.2222$$



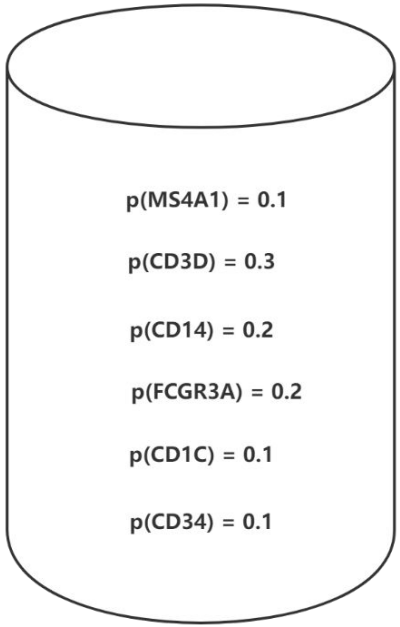


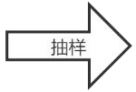
	MS4A1	CD3D	CD14	FCGR3A	CD1C	CD34
Barcode1	0	3	0	1	1	0
Barcode2	2	0	0	0	0	1
Barcode3	0	0	2	1	1	0



	MS4A1	CD3D	CD14	FCGR3A	CD1C	CD34
Barcode1	0	3	0	1	1	0
Barcode2	2	0	0	0	0	1
Barcode3	0	0	2	1	1	0

	Total count	pvalue
Barcode1	5	0.0107
Barcode2	3	0.0029
Barcode3	4	0.0096

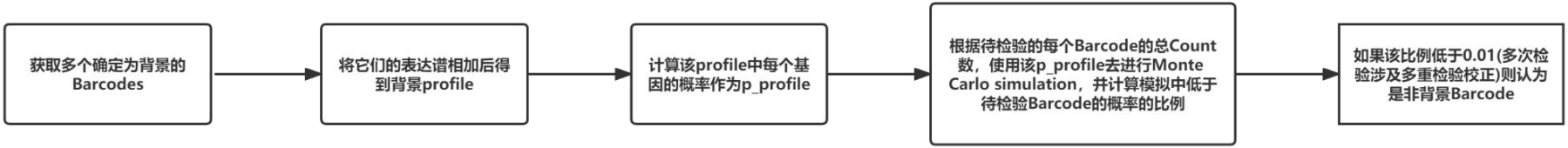




	MS4A1	CD3D	CD14	FCGR3A	CD1C	CD34
Barcode1	0	3	0	1	1	0
Barcode2	5	0	0	0	0	1
Barcode3	0	0	2	1	1	0

	Total count	pvalue	抽样1,000,000后小于该概率的比例	是否小概率事件(0.01)
Barcode1	5	0.0107	0.509838	否
Barcode2	6	6e-6	1.8e-05	是
Barcode3	4	0.0096	0.24403	否

从背景中分离出可能是细胞的Barcodes的思路



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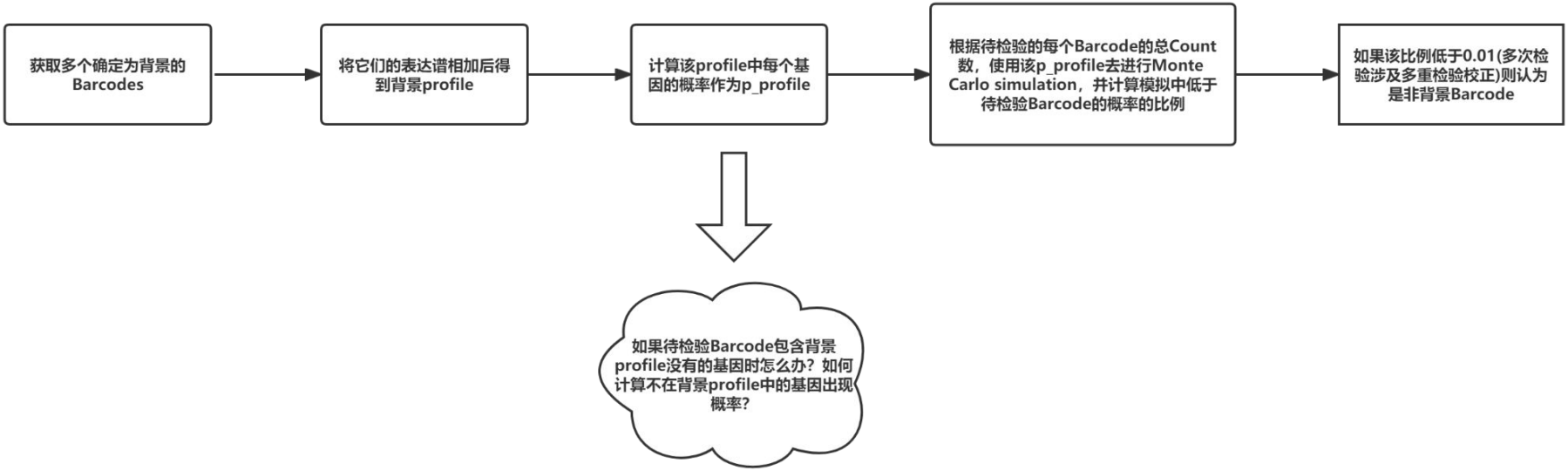

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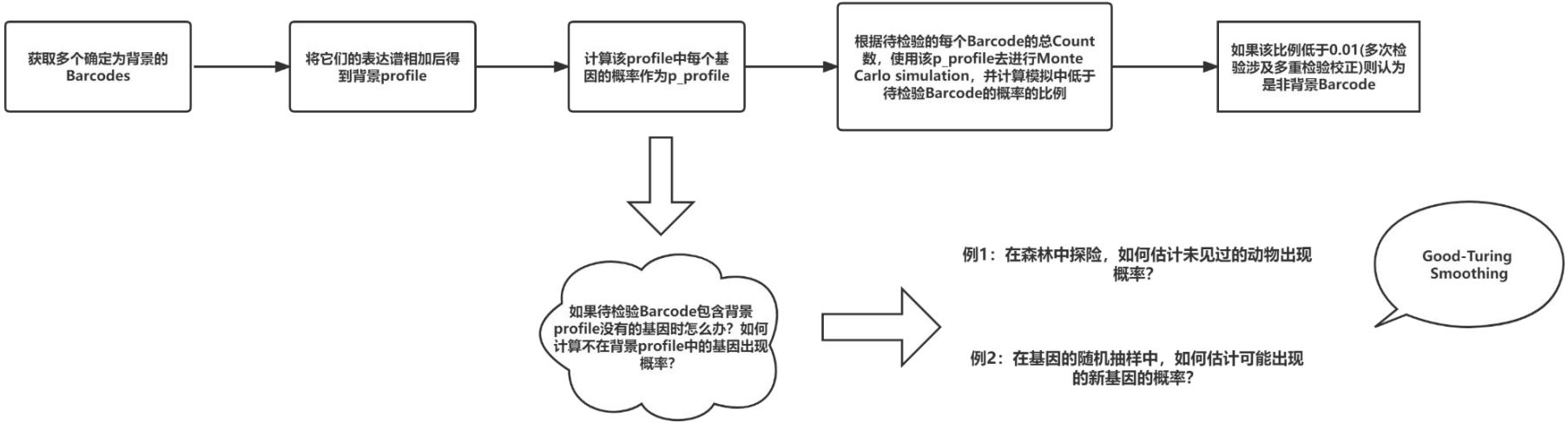


Good-Turing Smoothing

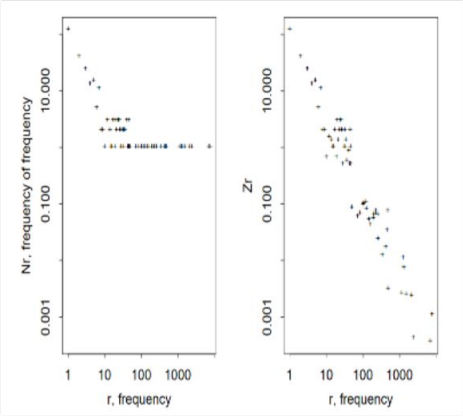
构建基因的频率及频率的频率列表

Frequency(r)	Frequency of Frequency(Nr)
1	100
2	10
3	3
4	0
5	1
...	...

从总的概率中分了一部分的概率到p0

$$p\mathbf{0} = p\mathbf{1} = \frac{N\mathbf{1}}{N}$$

smooth过程



将r ->r*

$$r^* = (r+\mathbf{1})\frac{S(N_r+\mathbf{1})}{S(N_r)}$$

$$p_r = r^*/N$$

Table 2
Partial Table of
Frequency of Frequencies for Prosody Example

frequency	frequency of frequency
r	N_r
1	120
2	40
3	24
4	13
5	15
6	5
7	11
8	2
9	2
10	1
11	0
12	3

$$p_0, rstar = SGT(freq, freqfreq)$$

这里的sum(rstar*freq)代表总Count数,pstar代表不同frequency的概率

$$pstar = (1 - p_0)(\frac{rstar}{sum(rstar * freq)})$$

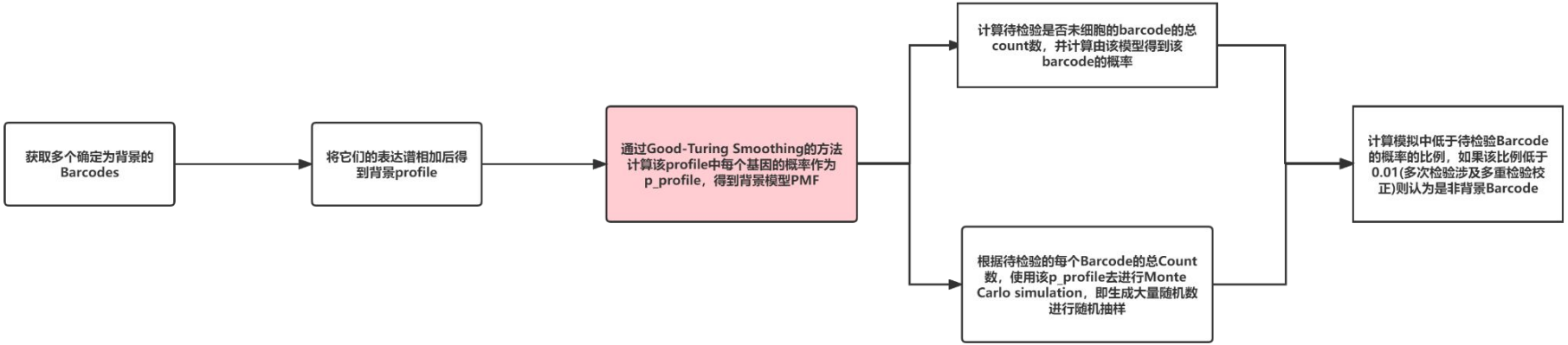
这里的p0代表遇到未出现过的基因的概率,p0i即是每一个新出现的基因的概率,n为所有可能出现的新基因的个数

$$p_{0i} = \frac{p_0}{n}$$

将p0i和pstar带入到背景barcode中得到ambient_profile_p

基因	MS4A1	CD14	CD3D	gene not in ambient_profile
count数	1	5	2	-
抽样概率	pstar(1)	pstar(5)	pstar(2)	p0i

从背景中分离出可能是细胞的Barcodes的思路



参考资料:

1. https://www.csie.ntu.edu.tw/~b92b02053/print/good-turing-smoothing-without.pdf
2. https://support.10xgenomics.com/single-cell-gene-expression/software/pipelines/latest/algorithms/overview
3. https://github.com/10XGenomics/cellranger/blob/a83c753ce641db6409a59ad817328354fbe7187e/lib/python/cellranger/cell_calling.py
4. https://github.com/10XGenomics/cellranger/blob/a83c753ce641db6409a59ad817328354fbe7187e/lib/python/cellranger/sgt.py
5. https://github.com/10XGenomics/cellranger/blob/a83c753ce641db6409a59ad817328354fbe7187e/lib/python/cellranger/stats.py